Contracts to Effects Compiler Typesoundness

Overview

The environment that will be used to type check target programs has the following Σ signature.

$$\Sigma(\mathtt{h}_{\mathscr{C}}) = \left\{\mathscr{C} : \forall\alpha.\langle \mathtt{blab}, \langle \alpha \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \mathsf{bool}, \alpha\rangle\rangle \to \alpha\right\}$$

$\cdot^*$: Contracts $\tau \to$ Effects τ

$$\begin{aligned}
\mathsf{num}^* &= \mathsf{num} \\
\mathsf{bool}^* &= \mathsf{bool} \\
\mathsf{unit}^* &= \mathsf{unit} \\
\langle \tau_1, \tau_2 \rangle^* &= \langle \tau_1^*, \tau_2^* \rangle \\
(\tau_1 \to \tau_2)^* &= \tau_1^* \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \tau_2^* \\
(x_1 : \tau_1, ..., x_n : \tau_n)^* &= x_1 : \tau_1^*, ..., x_n : \tau_n^*
\end{aligned}$$

Compiler Type Preservation

Theorem 0.1. (Compiler Type Preservation)

If $\Gamma \vdash e : \tau \rightarrowtail^d e'$ then $\Gamma^* \vdash e' : \tau^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$.

Proof.

By induction on structure of $\Gamma \vdash e : \tau \rightarrowtail^d e'$.

- Case $e = x$, $e = n$, $e = b$, $e = \mathsf{unit}$

  $e' = e$ — Given
  $\Gamma^* \vdash_{\mathrm{val}} e' : \tau^*$ — T-E-Var, T-Num, T-Bool, T-Unit
  $\Gamma^* \vdash e' : \tau^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ — T-E-Val

- Case

$$\frac{\Gamma, x : \tau_1 \vdash e : \tau_2 \rightarrowtail^d e''}{\Gamma \vdash \lambda x : \tau_1.e : \tau_1 \to \tau_2 \rightarrowtail^d \lambda x : \tau_1^*.e''}$$

  $\Gamma^*, x : \tau_1^* \vdash e'' : \tau_2^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ — I.H.
  $\Gamma^* \vdash_{\mathrm{val}} \lambda x : \tau_1^*.e'' : \tau_1^* \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \tau_2^*$ — T-E-Abs
  $\Gamma^* \vdash_{\mathrm{val}} \lambda x : \tau_1^*.e'' : (\tau_1 \to \tau_2)^*$ — Type compilation $\cdot^*$
  $\Gamma^* \vdash \lambda x : \tau_1^*.e'' : (\tau_1 \to \tau_2)^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ — T-E-Val

- Case

$$\frac{\Gamma \vdash e_1 : \tau_1 \to \tau_2 \rightarrowtail^d e_1' \qquad \Gamma \vdash e_2 : \tau_1 \rightarrowtail^d e_2'}{\Gamma \vdash e_1\ e_2 : \tau_2 \rightarrowtail^d e_1'\ e_2'}$$

$\Gamma^* \vdash e_1' : \tau_1^* \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \tau_2^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ I.H.
$\Gamma^* \vdash e_2' : \tau_1^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ I.H.
$\Gamma^* \vdash e_1'\ e_2' : \tau_2^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ T-E-App

The effect on the arrow and the effect of the premises are both $\langle \mathtt{h}_{\mathscr{C}} \rangle$.

- Case $\mu x : \tau.e$, $\langle e_1, e_2 \rangle$, $\mathsf{inj.l}(e)$, $\mathsf{inj.r}(e)$, if e_1 then e_2 else e_3, $o(e_1, ..., e_n)$

  $\Gamma^* \vdash e_i' : \tau_i^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ for every sub-expression I.H.
  $\Gamma^* \vdash e' : \tau^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ lifted rule of Base

  For $\mu x : \tau.e$, the binder is $x : \tau^*$, and the I.H. is used under $\Gamma^*, x : \tau^*$.

- Case

$$\frac{\Gamma \vdash e_3 : \tau \to \mathsf{bool} \rightarrowtail^d e_5 \qquad \Gamma \vdash e_4 : \tau \rightarrowtail^d e_6}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\mathsf{flat}(e_3), e_4) : \tau \rightarrowtail^d \mathtt{perform}\ \mathscr{C}\ \tau^*\ \langle k, \langle e_5, e_6 \rangle \rangle}$$

  $\Gamma^* \vdash e_5 : \tau^* \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \mathsf{bool} \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ I.H.
  $\Gamma^* \vdash e_6 : \tau^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ I.H.
  $\Gamma^* \vdash \mathtt{perform}\ \mathscr{C}\ \tau^*\ \langle k, \langle e_5, e_6 \rangle \rangle : \tau^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ Lemma 0.2

  This is the only rule that introduces $\mathtt{h}_{\mathscr{C}}$.

- Case

$$\frac{\Gamma \vdash e_1 : \tau_1 \to \tau_2 \rightarrowtail^d e_2 \qquad \Gamma \vdash \lambda f : \tau_1 \to \tau_2.\lambda x : \tau_1.\mathsf{mon}_j^{k,l}\left(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right) : (\tau_1 \to \tau_2) \to (\tau_1 \to \tau_2) \rightarrowtail^d e_3}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, e_1) : \tau_1 \to \tau_2 \rightarrowtail^d e_3\ e_2}$$

  $\Gamma^* \vdash e_2 : (\tau_1 \to \tau_2)^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ I.H.
  $\Gamma^* \vdash e_3 : ((\tau_1 \to \tau_2) \to (\tau_1 \to \tau_2))^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ I.H.
  $\Gamma^* \vdash e_3 : (\tau_1 \to \tau_2)^* \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} (\tau_1 \to \tau_2)^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ def. of $\cdot^*$
  $\Gamma^* \vdash e_3\ e_2 : (\tau_1 \to \tau_2)^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ T-E-App

- Case

$$\frac{\Gamma \vdash e_1 : \tau_1 \to \tau_2 \rightarrowtail^d e_2 \qquad \Gamma \vdash e_w : (\tau_1 \to \tau_2) \to (\tau_1 \to \tau_2) \rightarrowtail^d e_3}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y.\kappa_2, e_1) : \tau_1 \to \tau_2 \rightarrowtail^d e_3\ e_2}$$

  where $e_w = \lambda f : \tau_1 \to \tau_2.\lambda x : \tau_1.\mathsf{mon}_j^{k,l}\left(\kappa_2\left[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\right], f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right)$

  $\Gamma^* \vdash e_2 : (\tau_1 \to \tau_2)^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ I.H.
  $\Gamma^* \vdash e_3 : (\tau_1 \to \tau_2)^* \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} (\tau_1 \to \tau_2)^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ I.H., def. of $\cdot^*$
  $\Gamma^* \vdash e_3\ e_2 : (\tau_1 \to \tau_2)^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ T-E-App

The dependency y is replaced in the source by the indy monitor $\mathsf{mon}_j^{l,j}(\kappa_1, x) : \tau_1$, so the I.H. on e_w already covers κ_2 in the context $\Gamma, f : \tau_1 \to \tau_2, x : \tau_1$.

- Case

$$\frac{x \notin \mathrm{dom}(\Gamma) \quad \Gamma, x : \langle \tau_1, \tau_2 \rangle \vdash \mathsf{mon}_j^{k,l}(\kappa_1, \mathsf{inj.l}(x)) : \tau_1 \rightarrowtail^d e_1 \quad \Gamma, x : \langle \tau_1, \tau_2 \rangle \vdash \mathsf{mon}_j^{k,l}(\kappa_2, \mathsf{inj.r}(x)) : \tau_2 \rightarrowtail^d e_2 \quad \Gamma \vdash e : \langle \tau_1, \tau_2 \rangle \rightarrowtail^d e_3}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\langle \kappa_1, \kappa_2 \rangle, e) : \langle \tau_1, \tau_2 \rangle \rightarrowtail^d (\lambda x : \langle \tau_1, \tau_2 \rangle^* . \langle e_1, e_2 \rangle)\ e_3}$$

$\Gamma^*, x : \langle \tau_1, \tau_2 \rangle^* \vdash e_1 : \tau_1^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ — I.H.

$\Gamma^*, x : \langle \tau_1, \tau_2 \rangle^* \vdash e_2 : \tau_2^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ — I.H.

$\Gamma^*, x : \langle \tau_1, \tau_2 \rangle^* \vdash \langle e_1, e_2 \rangle : \langle \tau_1, \tau_2 \rangle^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ — T-Tuple

$\Gamma^* \vdash_{\mathrm{val}} \lambda x : \langle \tau_1, \tau_2 \rangle^* . \langle e_1, e_2 \rangle : \langle \tau_1, \tau_2 \rangle^* \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \langle \tau_1, \tau_2 \rangle^*$ — T-E-Abs

$\Gamma^* \vdash \lambda x : \langle \tau_1, \tau_2 \rangle^* . \langle e_1, e_2 \rangle : \langle \tau_1, \tau_2 \rangle^* \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \langle \tau_1, \tau_2 \rangle^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ — T-E-Val

$\Gamma^* \vdash e_3 : \langle \tau_1, \tau_2 \rangle^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ — I.H.

$\Gamma^* \vdash (\lambda x . \langle e_1, e_2 \rangle)\ e_3 : \langle \tau_1, \tau_2 \rangle^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ — T-E-App

□

Lemma 0.2. (Perform Type)

If $\Gamma \vdash e_1 : \tau \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \mathsf{bool} \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ and $\Gamma \vdash e_2 : \tau \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ then

$\Gamma \vdash \mathtt{perform}\ \mathscr{C}\ \tau\ \langle k, \langle e_1, e_2 \rangle \rangle : \tau \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$.

Proof.

$\Gamma \vdash_{\mathrm{val}} \mathtt{perform}\ \mathscr{C}\ \tau : \langle \mathtt{blab}, \langle \tau \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \mathsf{bool}, \tau \rangle \rangle \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \tau$ — T-E-Perform, $\alpha := \tau, \varepsilon := \langle\rangle$

$\Gamma \vdash \mathtt{perform}\ \mathscr{C}\ \tau : \langle \mathtt{blab}, \langle \tau \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \mathsf{bool}, \tau \rangle \rangle \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \tau \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ — T-E-Val

$\Gamma \vdash k : \mathtt{blab} \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ — T-E-Label

$\Gamma \vdash e_1 : \tau \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \mathsf{bool} \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ — Given

$\Gamma \vdash e_2 : \tau \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ — Given

$\Gamma \vdash \langle e_1, e_2 \rangle : \langle \tau \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \mathsf{bool}, \tau \rangle \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ — T-Tuple

$\Gamma \vdash \langle k, \langle e_1, e_2 \rangle \rangle : \langle \mathtt{blab}, \langle \tau \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \mathsf{bool}, \tau \rangle \rangle \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ — T-Tuple

$\Gamma \vdash \mathtt{perform}\ \mathscr{C}\ \tau\ \langle k, \langle e_1, e_2 \rangle \rangle : \tau \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$ — T-E-App

□

Wrapped Compiled Program Typing (Deep Handler)

The deep handler wrapper from the compiler equivalence proof removes $\mathtt{h}_{\mathscr{C}}$ from the effect row. We rewrite the wrapper as follows:

$$\mathsf{wrap}_d = \mu w : \forall\alpha.\big(\mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \alpha\big) \to^{\langle\rangle} \alpha.\Lambda\alpha.\lambda f : \mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \alpha.\mathtt{handler}_d(h_w)\ f$$

$$h_w = \big\{\mathscr{C} \to \Lambda\beta.\lambda\langle l, \langle e, v\rangle\rangle.\lambda k : \beta \to^{\langle\rangle} \alpha.\ \text{if } w[\mathsf{bool}]\ (\lambda_ : \mathsf{unit}.e\ v)\ \text{then } k\ v\ \text{else } \mathtt{error}_l\big\}$$

Note that h_w signature matches $\Sigma(\mathtt{h}_{\mathscr{C}})$ if we replace α with β.

Theorem 0.3. (Compiler Effect Safety with wrapper deep handler)

If $\cdot \vdash e : \tau \rightarrowtail^d e'$ then $\cdot \vdash \mathsf{wrap}_d[\tau^*]\ (\lambda_ : \mathsf{unit}.e') : \tau^* \mid \langle\rangle$.

Proof.

$\cdot \vdash e' : \tau^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$	Theorem 0.1
$\cdot \vdash_{\text{val}} \lambda_ : \mathsf{unit}.e' : \mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \tau^*$	T-E-Abs
$\cdot \vdash \mathsf{wrap}_d[\tau^*] : \big(\mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \tau^*\big) \to^{\langle\rangle} \tau^* \mid \langle\rangle$	Lemma 0.4, T-E-Tapp
$\cdot \vdash \mathsf{wrap}_d[\tau^*]\ (\lambda_ : \mathsf{unit}.e') : \tau^* \mid \langle\rangle$	T-E-App

□

Lemma 0.4. (Type of wrap deep handler)

$\cdot \vdash_{\text{val}} \mathsf{wrap}_d : \forall\alpha.\big(\mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \alpha\big) \to^{\langle\rangle} \alpha$

Proof.

Assume contexts $\Gamma_w = w : \forall\alpha.\big(\mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \alpha\big) \to^{\langle\rangle} \alpha$.

And $\Delta = \Gamma_w, l : \mathtt{blab}, e : \beta \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \mathsf{bool}, v : \beta, k : \beta \to^{\langle\rangle} \alpha$.

$\Delta \vdash e\ v : \mathsf{bool} \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$	T-E-App
$\Delta \vdash_{\text{val}} \lambda_ : \mathsf{unit}.e\ v : \mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \mathsf{bool}$	T-E-Abs
$\Delta \vdash w[\mathsf{bool}] : \big(\mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \mathsf{bool}\big) \to^{\langle\rangle} \mathsf{bool} \mid \langle\rangle$	T-E-Tapp
$\Delta \vdash w[\mathsf{bool}]\ (\lambda_ : \mathsf{unit}.e\ v) : \mathsf{bool} \mid \langle\rangle$	T-E-App
$\Delta \vdash k\ v : \alpha \mid \langle\rangle$	T-E-App
$\Delta \vdash \mathtt{error}_l : \alpha \mid \langle\rangle$	T-E-Error
$\Delta \vdash \text{if } w[\mathsf{bool}]\ (\lambda_ : \mathsf{unit}.e\ v)\ \text{then } k\ v\ \text{else } \mathtt{error}_l : \alpha \mid \langle\rangle$	T-If
$\Gamma_w \vdash_{\text{val}} h_{w(\mathscr{C})} : \forall\beta.\langle \mathtt{blab}, \langle \beta \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \mathsf{bool}, \beta\rangle\rangle \to^{\langle\rangle} \big(\big(\beta \to^{\langle\rangle} \alpha\big) \to^{\langle\rangle} \alpha\big)$	T-E-Abs, T-E-TAbs
$\Gamma_w \vdash_{\text{ops}} h_w : \alpha \mid \langle \mathtt{h}_{\mathscr{C}} \mid \langle\rangle\rangle$	T-E-Ops
$\Gamma_w \vdash_{\text{val}} \mathtt{handler}_d(h_w) : \big(\mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \alpha\big) \to^{\langle\rangle} \alpha$	T-E-Handler
$\Gamma_w, f : \mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \alpha \vdash \mathtt{handler}_d(h_w)\ f : \alpha \mid \langle\rangle$	T-E-App
$\Gamma_w \vdash_{\text{val}} \lambda f : \mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \alpha.\mathtt{handler}_d(h_w)\ f : \big(\mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \alpha\big) \to^{\langle\rangle} \alpha$	T-E-Abs
$\Gamma_w \vdash_{\text{val}} \Lambda\alpha.\lambda f.\mathtt{handler}_d(h_w)\ f : \forall\alpha.\big(\mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \alpha\big) \to^{\langle\rangle} \alpha$	T-E-TAbs
$\cdot \vdash_{\text{val}} \mathsf{wrap}_d : \forall\alpha.\big(\mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \alpha\big) \to^{\langle\rangle} \alpha$	T-Fix

□

Wrapped Compiled Program Typing (Shallow Handler)

$$\mathsf{wrap} = \Lambda\alpha.\, \lambda f : \mathsf{unit} \to \alpha.\, \mathsf{H}\ f$$

$$\mathsf{H} = \mu h.\mathtt{handler}_s(\{\mathscr{C} \to \lambda\langle l, \langle e, v\rangle\rangle.\lambda k.h\ (\lambda_.\ \text{if } e\ v \text{ then } k\ v \text{ else } \mathtt{error}_l)\})$$

Note that h_w signature matches $\Sigma(\mathtt{h}_{\mathscr{C}})$ if we replace α with β.

Theorem 0.5. (Compiler Effect Safety with wrapper shallow handler)

If $\cdot \vdash e : \tau \rightarrowtail^s e'$ then $\cdot \vdash \mathsf{wrap}_s[\tau^*]\ (\lambda_ : \mathsf{unit}.e') : \tau^* \mid \langle\rangle$.

Proof.

$\cdot \vdash e' : \tau^* \mid \langle \mathtt{h}_{\mathscr{C}} \rangle$	Theorem 0.1
$\cdot \vdash_{\text{val}} \lambda_ : \mathsf{unit}.e' : \mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \tau^*$	T-E-Abs
$\cdot \vdash \mathsf{wrap}_s[\tau^*] : \left(\mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \tau^*\right) \to^{\langle\rangle} \tau^* \mid \langle\rangle$	Lemma 0.6, T-E-Tapp
$\cdot \vdash \mathsf{wrap}_s[\tau^*]\ (\lambda_ : \mathsf{unit}.e') : \tau^* \mid \langle\rangle$	T-E-App

□

Lemma 0.6. (Type of wrap shallow handler)

$\cdot \vdash_{\text{val}} \mathtt{wrap}_s : \forall\alpha.\left(\mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \alpha\right) \to^{\langle\rangle} \alpha$.

Proof.

Assume $\Gamma_w = h : \forall\alpha.\left(\mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \alpha\right) \to^{\langle\rangle} \alpha$.

$\Gamma_w \vdash_{\text{val}} \mathsf{H} : \forall\beta.\langle \mathtt{blab}, \langle \beta \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \mathsf{bool}, \beta\rangle\rangle \to^{\langle\rangle} \left(\left(\beta \to^{\langle\rangle} \alpha\right) \to^{\langle\rangle} \alpha\right)$	T-E-Abs, T-E-TAbs
$\Gamma_w \vdash_{\text{ops}} \mathsf{H} : \alpha \mid \langle \mathtt{h}_{\mathscr{C}} \mid \langle\rangle\rangle$	T-E-Ops
$\Gamma_w \vdash_{\text{val}} \mathtt{handler}_s(\mathsf{H}) : \left(\mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \alpha\right) \to^{\langle\rangle} \alpha$	T-E-Handler
$\Gamma_w \vdash_{\text{val}} \lambda f : \mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \alpha.\mathtt{handler}_s(\mathsf{H})\ f : \left(\mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \alpha\right) \to^{\langle\rangle} \alpha$	T-E-Abs
$\cdot \vdash_{\text{val}} \Lambda\alpha.\lambda f.\mathtt{handler}_s(\mathsf{H})\ f : \forall\alpha.\left(\mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \alpha\right) \to^{\langle\rangle} \alpha$	T-E-TAbs
$\cdot \vdash_{\text{val}} \mathtt{wrap}_s : \forall\alpha.\left(\mathsf{unit} \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \alpha\right) \to^{\langle\rangle} \alpha$	T-Fix

□